# JAHNAVI KACHHIA

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## SUMMARY

AI and Machine Learning professional with 7+ years of experience designing, developing, and deploying scalable, production-grade systems across healthcare, technology, and enterprise domains. Experienced in end-to-end ML product ownership, covering data strategy, model optimization, and deployment at scale. Skilled in Generative AI (RAG, LLM fine-tuning), predictive modeling, and multimodal learning, with strong expertise in MLOps, distributed systems. Passionate about building responsible, human-centered AI that drives innovation and measurable impact.

## WORK EXPERIENCE

**Global Product Owner, AI & ML**
*Abbott* **11/2025- Present**

- Orchestrated data and ML workflows with **Apache Airflow, AWS Glue,** and **Athena**, enabling automated data preprocessing and real-time feature extraction for AI models.
- Utilize **Retrieval-Augmented Generation (RAG)** techniques to enhance response quality and context-awareness in **Generative AI** Models, boosting performance in complex tasks.
- **Lead global AI and ML initiatives** for **Abbott's FreeStyle Libre** continuous glucose monitoring (CGM) platform, used by **5M+ users worldwide**, driving advancements in **Predictive Analytics**, **Personalization**, and **Decision Intelligence** to improve patient outcomes.

**Senior Machine Learning Engineer** **09/2025 - 11/2025**
*Accompany Health*

- Built and productionized scalable ML pipelines using **Amazon SageMaker Pipelines**, integrating model training, evaluation, and deployment workflows within a CI/CD framework.
- Designed and implemented **feature stores and model monitoring systems** (e.g., for drift detection and data quality checks), ensuring ongoing model performance and accountability in production environments.

**ML Software Engineer** **04/2024 - 04/2025**
*Meta*

- **Leverage Multi-Modal Large Language Models (LLMs)** and **Prompt Engineering** to optimize system performance across various use cases, enhancing model capabilities for complex data integration.
- Develop and Implement **Re-Ranking Strategies**, improving the accuracy of search results by **6%**, optimizing model outputs for higher relevance and precision.
- Build **Responsible AI** by integrating **Ethical Considerations**, **Bias Mitigation Strategies**, and ensuring **Fairness in NLP Models**, particularly for sensitive and high-stakes applications.
- Experienced in **fine-tuning large language models (LLMs)** for specific tasks using frameworks like **TensorFlow and PyTorch**, and deploying models at scale with **Kubernetes**.

**Machine Learning Engineer** **04/2021 - 04/2024**
*Tata Consultancy Services - End Client Apple*

- Integrated **NLP based automated question-answer system** in a slack channel that increased users satisfaction by **15%** resulting in **20%** more traffic handling capacity.
- Data mining HDFS files and performed data cleaning and feature selection using MLLib package in **PySpark**
- Extensively working on **Natural Language Processing (NLP)** libraries such as **Spacy, NLTK,Gensim, Flair, and sklearn-crfsuite**

- Successfully developed and implemented NLP solutions for a variety of tasks such as **Entity Recognition and Extraction, Sentiment Analysis, Text Summarization and Machine Translation**.
- Deployed and monitored machine learning models in production using **AWS, Docker and Kubernetes**

**Data Analyst** **02/2020 - 03/2021**
*Data Application Lab*
- Transformed complex data using **Python, SQL, Tableau, Spark** to generate actionable insights.
- Designed models with **88% accuracy** to detect credit card defaults, reducing bank expenses by **23%**.
- Gained hands-on experience with **AWS (EC2, S3, EMR)** and **Google Cloud** for provisioning virtual clusters.

**Machine Learning Researcher** **05/2019 - 05/2020**
*California State University Fullerton*
- **Brain-Computer Interfaces (BCI) with ML**: Classified primitive shapes from EEG signals and converted them into CAD files, enabling an **automatic 3D printer** to create shapes based on human thoughts.
- **Deep Learning Expertise**: Implemented models like **CNN**, **RNN**, and **LSTM** using **TensorFlow** and **Keras**.

## EDUCATION

**Master of Science in Computer Engineering** **08/2018 - 05/2020**
*California State University Fullerton* GPA 3.9
**Machine Learning Professional** **10/2019 - 03/2020**
*Stanford University* Certified
**Bachelor of Engineering in Electronics and Communication** **08/2009 - 05/2013**
*Madhuben and Bhanubhai Patel Women's Inst. Of Eng.* GPA 3.2

## SKILLS

**Programming Languages:** Python, PostgreSQL, PySpark, MATLAB, C++, Tableau
**Soft Skills:** Problem-solving, Critical thinking, Communication and Time-management
**Technical Specialty:** Docker, Kubernetes, Git, AWS, Google Cloud, Spark, Hadoop, Excel, PowerPoint
**ML Libraries:** NLTK, SpaCy, Flair, SciPy, Keras, Matplotlib, OpenCV, Pandas, PyTorch, Seaborn, TensorFlow
**Mathematics:** Linear Algebra, Matrices, Calculus, Probability, and Statistics

## CERTIFICATES

- Fundamentals of **Deep Learning for Computer Vision** certified by **NVIDIA**
- **Getting Started with AWS Machine Learning** certified by **AWS**

## RESEARCH PUBLICATIONS

- "**EEG-based Image Classification using Machine Learning Algorithms**", 2021 10th Annual Computing and Communication Workshop and Conference (CCWC), Las Vegas, NV, USA, 2021
- "**Deep Learning Enhanced BCI Technology for 3D Printing**," 2020 11th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York, NY, USA, 2020
- "**Automated Radar Signal Analysis Based on Deep Learning**", 2020 10th Annual Computing and Communication Workshop and Conference (CCWC), Las Vegas, NV, USA, 2020